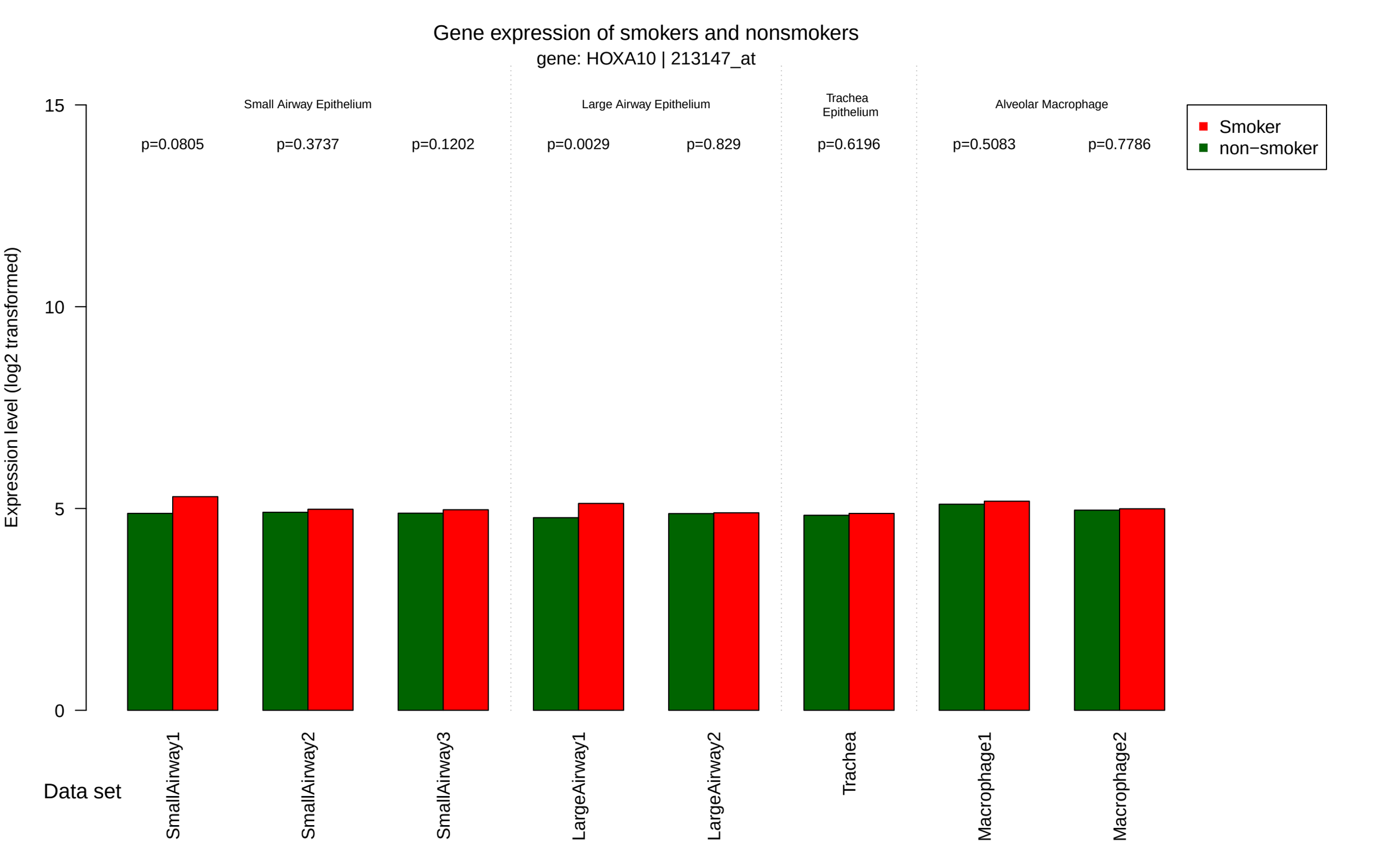
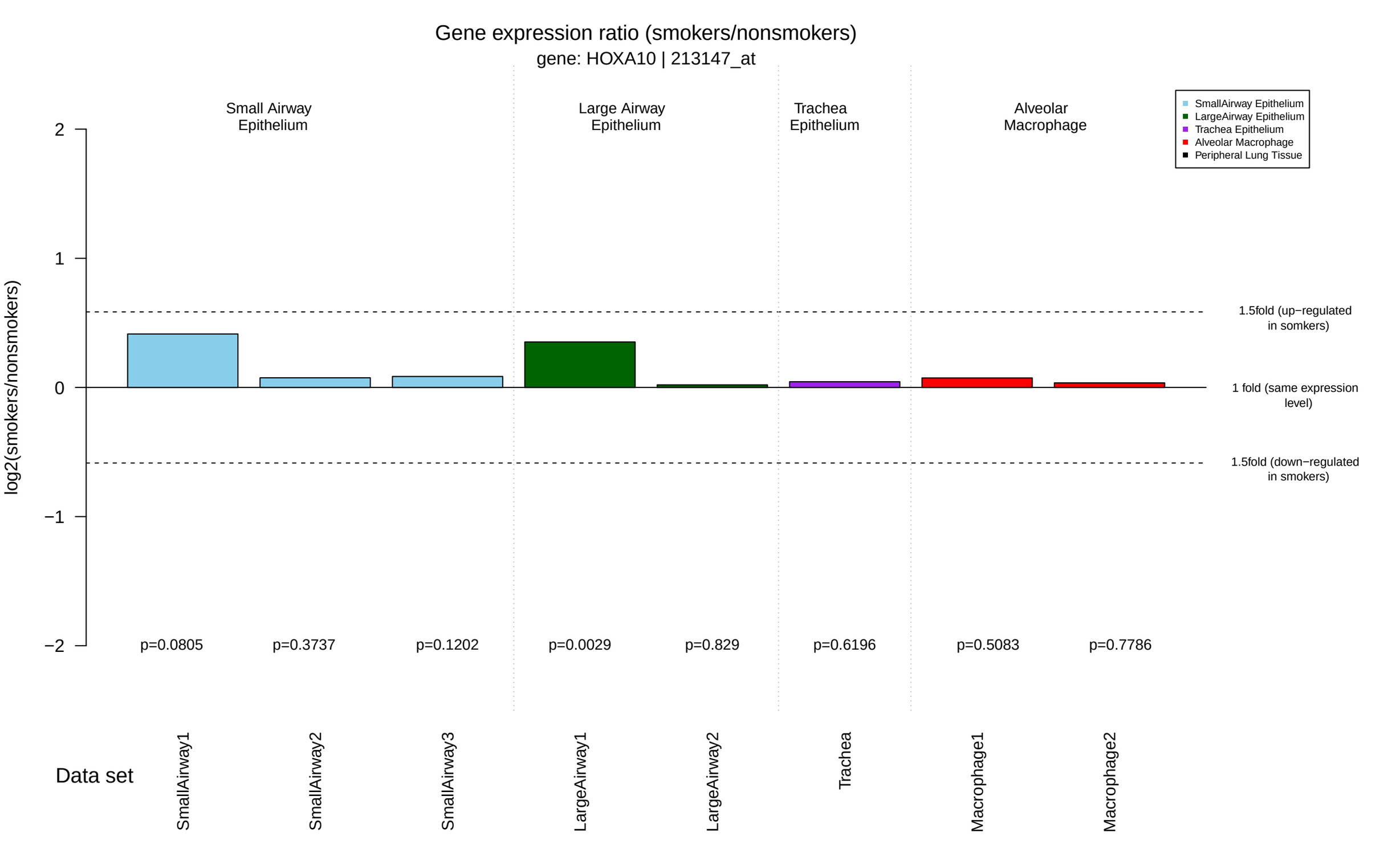
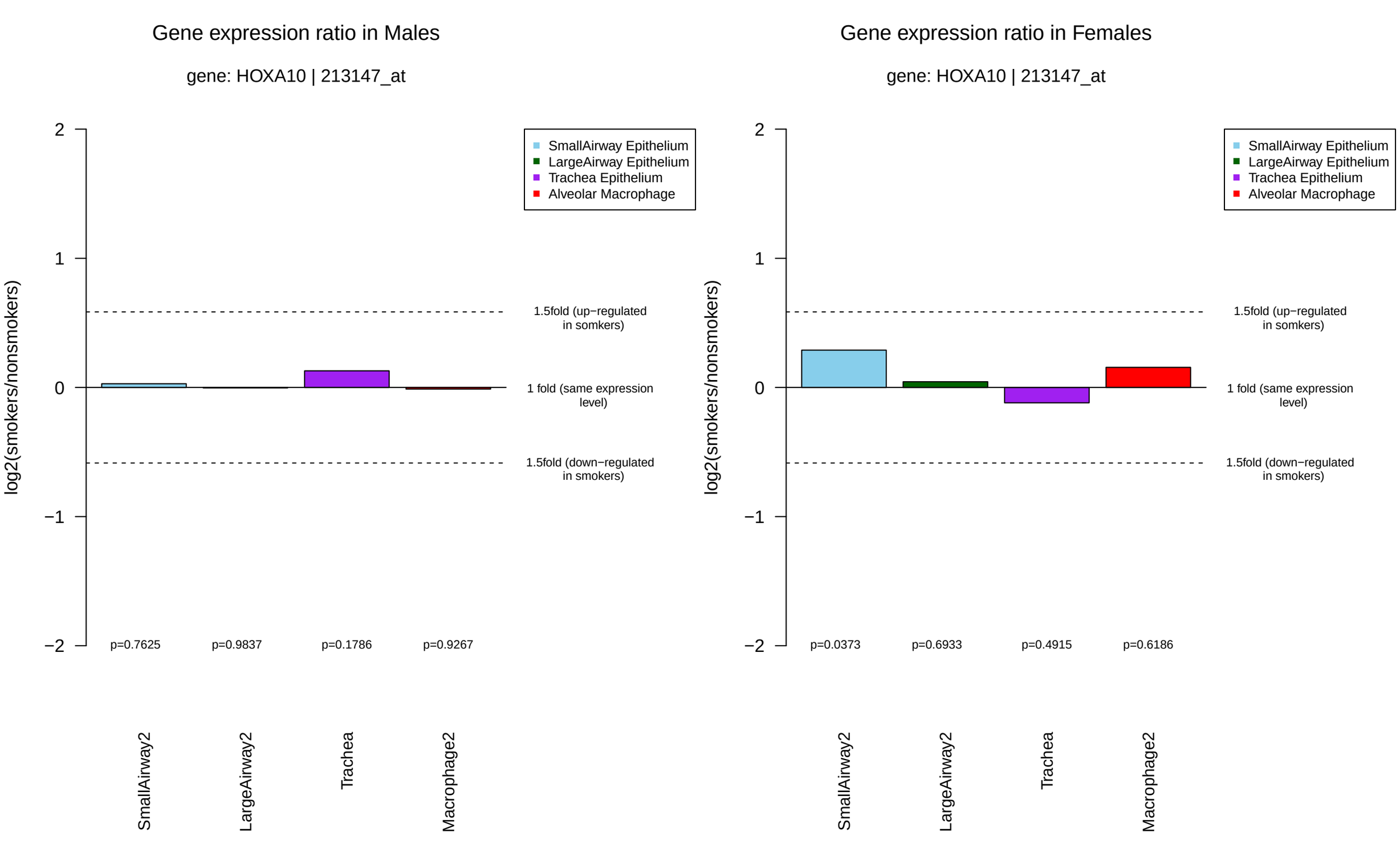
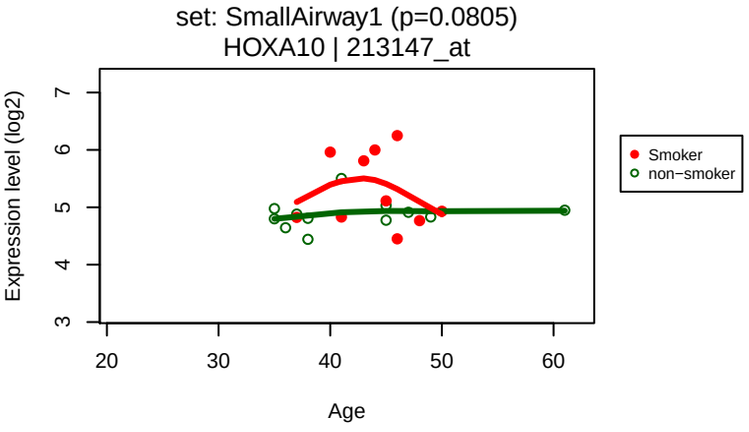
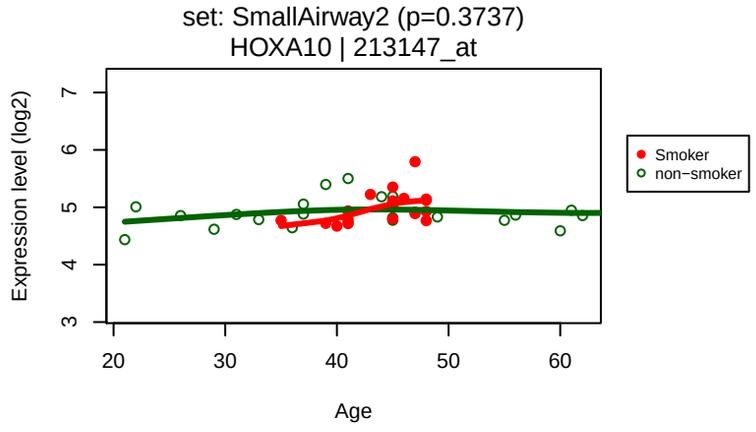
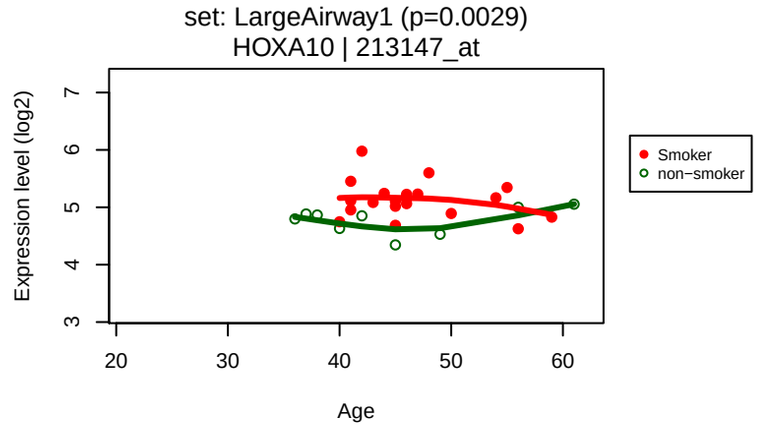

Gene expression of smokers and nonsmokers
gene: HOXA10 | 213147_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0805
p=0.3737
p=0.1202
p=0.0029
p=0.829
p=0.6196
p=0.5083
p=0.7786
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

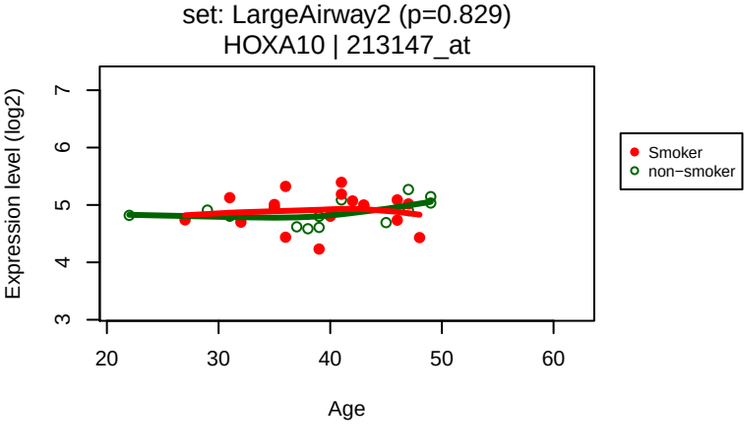

Gene expression ratio (smokers/nonsmokers)
gene: HOXA10 | 213147_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0805
p=0.3737
p=0.1202
p=0.0029
p=0.829
p=0.6196
p=0.5083
p=0.7786
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

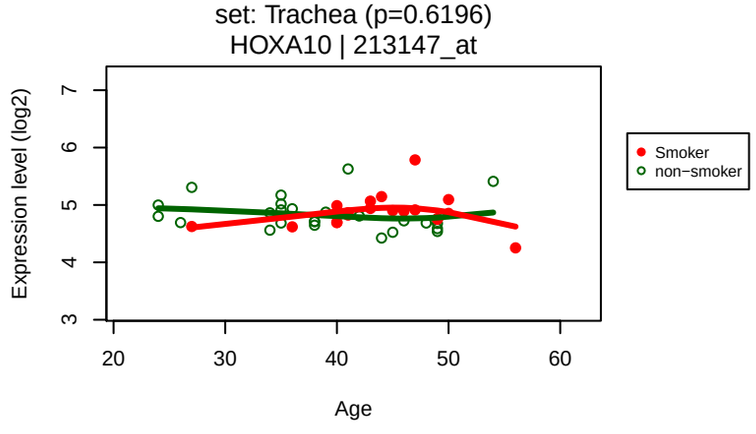

Gene expression ratio in Males
gene: HOXA10 | 213147_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7625
p=0.9837
p=0.1786
p=0.9267
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA10 | 213147_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0373
p=0.6933
p=0.4915
p=0.6186
SmallAirway2
LargeAirway2
Trachea
Macrophage2

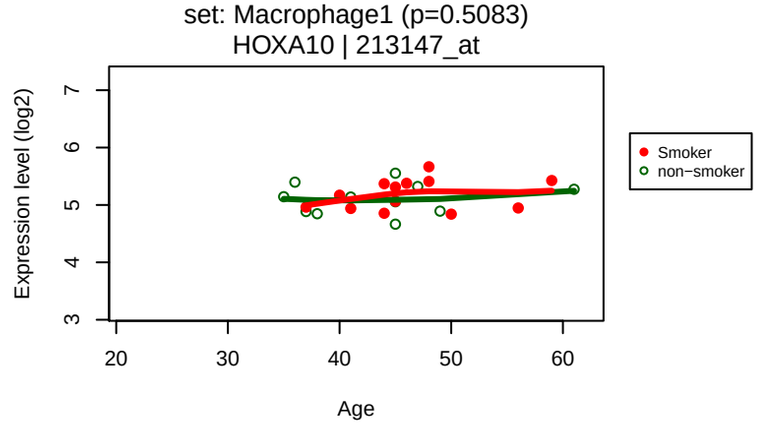

set: SmallAirway1 (p=0.0805)
HOXA10 | 213147_at
Expression level (log2)
Age
Smoker
non−smoker

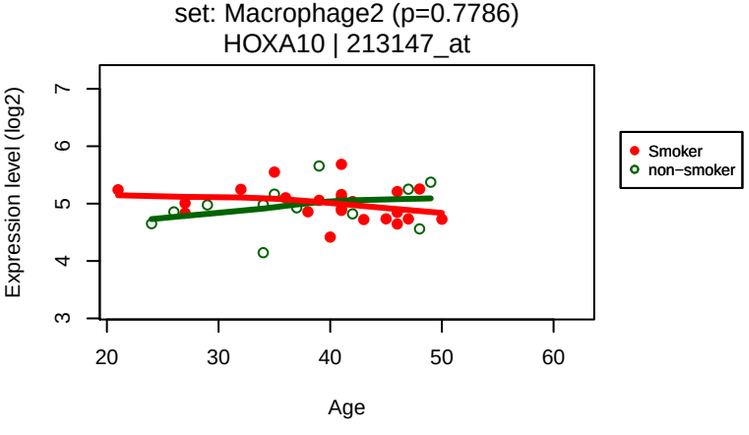

set: SmallAirway2 (p=0.3737)
HOXA10 | 213147_at
Expression level (log2)
Age
Smoker
non−smoker

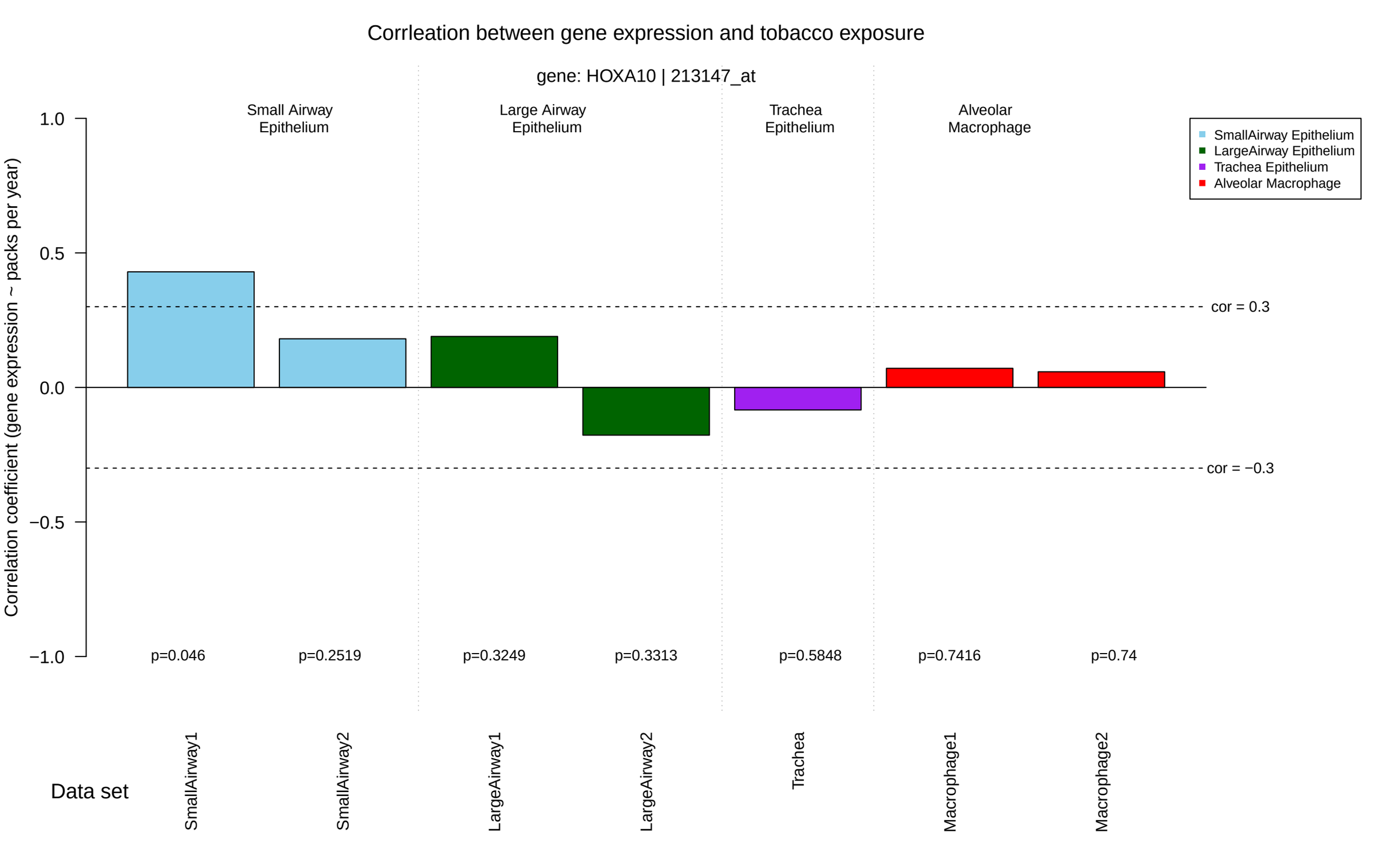

set: LargeAirway1 (p=0.0029)
HOXA10 | 213147_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.829)
HOXA10 | 213147_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6196)
HOXA10 | 213147_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5083)
HOXA10 | 213147_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7786)
HOXA10 | 213147_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA10 | 213147_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.046
p=0.2519
p=0.3249
p=0.3313
p=0.5848
p=0.7416
p=0.74
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2